

Create a large file of a specific size quickly

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Quickly creates a file of a specific size (e.g., 1 GB) using `fallocate`, which is much faster than `dd` because it allocates blocks without writing actual data.

Language

bash

Category

filesystem

Command

```
fallocate -l 1G test_file
```

Explanation

The `'fallocate'` command communicates directly with the filesystem to preallocate disk space for a file. Unlike `'dd'`, which must write every byte to disk, `'fallocate'` simply marks the space as allocated in the filesystem's metadata. This is extremely efficient for creating large test files, swap files, or preallocating space for databases. The length `(-l)` can be specified in bytes or using suffixes like K, M, G, T, P, E.

Tags

fallocate, filesystem, storage, disk-usage, allocation, testing

Dependencies

util-linux

Arguments

1. **SIZE** (Optional): The size of the file to create (e.g., 10M, 1G, 500M).
Default: 1G
2. **FILENAME** (Optional): The name of the file to be created.
Default: test_file

Examples

1. `fallocate -l 100M dummy_file` - Create a 100 MB file named `dummy_file`
2. `fallocate -l 1G large_test_file` - Create a 1 GB file for testing
3. `fallocate -l 10G /swapfile` - Preallocate 10 GB for a swap file (requires subsequent `mkswap` and `swapon`)

Output

(No output on success)

Notes

- Not all filesystems support `fallocate`. It is well-supported on `ext4`, `xfs`, `btrfs`, and `ocfs2`.
- If the filesystem doesn't support `fallocate`, you can use `'truncate -s SIZE FILENAME'` as a fallback, though it creates a sparse file.
- For older systems or filesystems, `'dd if=/dev/zero of=FILENAME bs=1M count=SIZE_IN_MB'` is the most compatible but much slower method.

Warnings

- This command will overwrite the specified file if it already exists.
- On some filesystems, the allocated space may contain 'stale' data from previously deleted files, though modern Linux filesystems (`ext4`, `xfs`) zero-fill or mark it as uninitialized for security.
- Ensure you have enough free disk space before running this command.

See Also

- `find-large-files-recursive`
- `disk-space-usage-per-directory`

Status

reviewed

Safety

caution

Shell

bash

Platforms

linux

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